

JDScience 11+ Mathematics Practice Paper 1 — Answers and Explanations

Mathematics · Answers and Explanations

Name _____

Date _____

Score _____

Recommended time: —

Questions: 40

Calculator: Not required unless a question says otherwise

Instructions

1. Issue this document separately from the pupil paper.
2. Each explanation shows the correct option and the reasoning.

This is an original JDScience practice paper written for school entrance exam preparation. It is not an official GL Assessment, CEM, CSSE, school or publisher paper.

Mark scheme

1. B — Units: $7 + 6 = 13$ (write 3, carry 1). Tens: $8 + 9 + 1 = 18$ (write 8, carry 1). Hundreds: $4 + 5 + 1 = 10$ (write 0, carry 1). Thousands: $3 + 2 + 1 = 6$. So $3\,487 + 2\,596 = 6\,083$.

2. B — Working column by column with borrowing gives $8\,052 - 3\,679 = 4\,373$.

3. C — $47 \times 30 = 1\,410$ and $47 \times 6 = 282$. Total: $1\,410 + 282 = 1\,692$.

4. B — $12 \times 200 = 2\,400$. Remainder $48 \div 12 = 4$. So $2\,448 \div 12 = 204$.

5. B — The lowest common multiple of 8 and 12 is 24. Of the options, only 24 is divisible by both.

6. D — Common factors of 36 and 54 include 1, 2, 3, 6, 9 and 18. The highest is 18.

7. A — The hundreds digit is 3, which is less than 5, so round down to 47 000.

8. B — $1/5$ of 85 = 17, so $3/5$ of 85 = $3 \times 17 = 51$.

9. B — $2\frac{1}{4} = 2\frac{2}{8}$. Then $2\frac{2}{8} + 1\frac{3}{8} = 3\frac{5}{8}$.

10. A — Divide numerator and denominator by 14: $42 \div 14 = 3$ and $56 \div 14 = 4$, so $3/4$.

Mark scheme

11. B — $0.75 = 75/100$. Divide by 25 to get $3/4$.

12. B — $46 \times 3 = 138$ with two decimal places in total, giving 1.38.

13. C — Multiply both numbers by 10: $72 \div 8 = 9$.

14. B — 10% of £60 = £6, so 30% = £18 and 5% = £3. Total £21.

15. B — 15% of £40 = £6. Sale price = £40 – £6 = £34.

16. D — Profit = £15. Percentage profit = $(15 \div 25) \times 100 = 60\%$.

17. C — Total parts = 9. One part = £8. Larger share = $7 \times £8 = £56$.

18. B — Per person: 75 g. For 10 people: 750 g.

19. B — $4 \times 50\,000 = 200\,000\text{ cm} = 2\,000\text{ m} = 2\text{ km}$.

20. B — Perimeter = $2 \times (12 + 7) = 38\text{ cm}$.

Mark scheme

21. B — Side length = $\sqrt{81} = 9$ cm.

22. A — Triangle area = $\frac{1}{2} \times 6 \times 4 = 12$ cm². Volume = $12 \times 10 = 120$ cm³.

23. B — $180^\circ - 47^\circ - 68^\circ = 65^\circ$.

24. B — Interior angle sum = $(6 - 2) \times 180^\circ = 720^\circ$. Each angle = $720^\circ \div 6 = 120^\circ$.

25. B — From 09:45 to 12:00 is 2 h 15 min; plus 20 min to 12:20 gives 2 h 35 min.

26. C — $14:30 + 1$ hour = 15:30; then + 50 minutes = 16:20.

27. B — $3.65 \times 1\,000 = 3\,650$ g.

28. C — $1.2 \times 1\,000 = 1\,200$ ml.

29. A — $3 \times \text{£}1.85 = \text{£}5.55$. Change = $\text{£}10.00 - \text{£}5.55 = \text{£}4.45$.

30. B — Cost per apple = $\text{£}0.40$. Fifteen apples cost $\text{£}6.00$.

Mark scheme

31. C — Total = $5 \times 12 = 60$. Known sum = 47. Fifth number = $60 - 47 = 13$.

32. B — Cyclists = $30 - 12 - 10 = 8$. Fraction = $8/30 = 4/15$.

33. C — Ordered values: 3, 4, 6, 8, 9. The middle value is 6.

34. B — Blue sections = 2 out of 8, so probability = $2/8 = 1/4$.

35. B — Outer area = $18 \times 12 = 216 \text{ m}^2$. Inner rectangle = $16 \times 10 = 160 \text{ m}^2$. Path area = $216 - 160 = 56 \text{ m}^2$.

36. B — After the book: $\text{£}48 - \text{£}16 = \text{£}32$. Snack costs $\frac{1}{4}$ of $\text{£}32 = \text{£}8$. Left: $\text{£}32 - \text{£}8 = \text{£}24$.

37. C — Speed = $180 \div 3 = 60 \text{ km/h}$. Distance in 5 hours = $60 \times 5 = 300 \text{ km}$.

38. C — Each term is double the previous plus 1: $31 \times 2 + 1 = 63$.

39. D — One face = $5 \times 5 = 25 \text{ cm}^2$. Six faces = 150 cm^2 .

40. D — $\text{£}50 \div \text{£}3.50 = 14.285\dots$ so she needs 15 full weeks ($15 \times \text{£}3.50 = \text{£}52.50$).